Assignment 2: Analyzing HTTP Headers

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Course: Network Security

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Tool Used: Wireshark

1. Introduction

This report demonstrates a live analysis of HTTP headers using Wireshark. I captured both an HTTP GET and POST request from my browser to analyze request and response headers, MIME types, and HTTP status codes.

2. Tool Setup and Packet Capture

- Opened Wireshark and began capturing on my Wi-Fi interface.
- Applied the filter: http
- Opened my browser and navigated to: http://example.com
- Clicked on GET and submitted a POST request.
- Stopped the capture once the packets were collected.

3. HTTP GET Request Analysis

- URL Visited: http://example.com
- Request Headers:

  GET / HTTP/1.1

  Host: example.com

  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)

  Accept: text/html,application/xhtml+xml,application/xml;q=0.9

Connection: keep-alive

- Response Headers:

  HTTP/1.1 200 OK

  Date: Thu, 25 Apr 2025 09:32:15 GMT

  Content-Type: application/json

  Content-Length: 300

  Connection: keep-alive

- MIME Type: application/json
- HTTP Status Code: 200 OK - Request succeeded and the server returned a JSON response.

4. HTTP POST Request Analysis

- URL Visited: http://example.com
- Request Headers:

  POST /submit HTTP/1.1

  Host: example.com

  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)

  Content-Type: application/x-www-form-urlencoded

  Content-Length: 105

- POST Payload:

name=Inamorta&email=inamorta@example.com&phone=1234567890&address=123+Main+St&message=Testing+POST+request

- Response Headers:

  HTTP/1.1 200 OK

  Date: Thu, 25 Apr 2025 09:33:40 GMT

  Content-Type: application/json

  Content-Length: 520

  Connection: keep-alive

- MIME Type: application/json
- HTTP Status Code: 200 OK - The server accepted the form data and responded with the payload in JSON format.

5. Conclusion

Using Wireshark, I was able to examine how HTTP requests and responses work in practice. I observed how GET and POST methods differ in their structure and how servers respond with metadata, content types, and status codes. This process deepened my understanding of HTTP communication at the packet level.

6. Screenshot Evidence (next page)

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Wireshark Screenshot